MODEL QUESTION PAPER – END SEMESTER EXAMINATION

**B.E. / B.Tech. – CSE (Artificial Intelligence & Machine Learning)**

**22CAC09N – FULL STACK WEB DEVELOPMENT**

**Time:** 3 Hours **Maximum Marks:** 60

SECTION – A

Answer ALL questions. **5 × 2 = 10 Marks**

| Q. No. | Question | CO | BL |
|---|---|---|---|
| 1 | Differentiate between  and  tags in HTML5. | CO1 | 2 |
| 2 | What is AJAX? Mention any two advantages. | CO2 | 2 |
| 3 | Differentiate between state and props in React. | CO3 | 2 |
| 4 | What is Node.js? | CO4 | 2 |
| 5 | Differentiate between GET and POST requests in Express.js. | CO5 | 2 |

SECTION – B

Answer ONE question from each Unit. 5 × 10 = 50 Marks

| Q. No. | Question | CO | BL |
|---|---|---|---|
| 6. (a) | Explain the structure of an HTML5 document. Discuss the purpose of headings, paragraphs, links, images, tables, lists, forms, semantic elements, , and  with suitable examples. | CO1 | BL2 |
| 6. (b) | Design a responsive personal portfolio webpage using HTML5 and CSS3. Explain the application of CSS selectors, Box Model, CSS Grid, Flexbox, margins, padding, borders, and positioning. | CO1 | BL3 |
| | OR | | |
| 7. (a) | Explain the important features of Bootstrap 5. Discuss the usage of containers, grid system, typography, buttons, cards, navigation bars, forms, and responsive utilities with examples. | CO1 | BL2 |
| 7. (b) | Develop a responsive college events webpage using Bootstrap 5. The webpage should include a navigation bar, event cards, registration form, buttons, and a responsive layout. | CO1 | BL3 |
| 8. (a) | Explain JavaScript data types, variables, operators, type conversion, functions, arrays, objects, and event handling with suitable examples. | CO2 | BL2 |
| 8. (b) | Develop a JavaScript-based online quiz application containing 10 multiple-choice questions. Explain the use of DOM manipulation, event handling, timer, answer validation, and score calculation. | CO2 | BL3 |
| | OR | | |
| 9. (a) | Explain AJAX and asynchronous communication in web applications. Compare XMLHttpRequest and Fetch API. Discuss JSON data exchange with suitable examples. | CO2 | BL2 |
| 9. (b) | Develop a weather dashboard using JavaScript Fetch API. Explain how data is | CO2 | BL3 |

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|---|---|---|---|
| | requested from an API, converted from JSON, displayed dynamically, and handled in case of errors. | | |
| 10. (a) | Explain JSX, functional components, props, state, event handling, Virtual DOM, and lifecycle-related concepts in React. Demonstrate the use of the useState() and useEffect() hooks with examples. | CO3 | BL2 |
| 10. (b) | Develop a React-based Student Management System using functional components, props, state, event handling, and list rendering. The application should allow adding, displaying, and deleting student records. | CO3 | BL3 |
| | OR | | |
| 11. (a) | Explain React Router and its importance in developing single-page applications. Develop a React application containing Home, About, Events, and Contact pages with navigation links and active-route highlighting. | CO3 | BL3 |
| 11. (b) | Explain Angular and Angular CLI. Discuss the creation and management of Angular components and templates. Demonstrate property binding and event binding with suitable examples. | CO3 | BL2 |
| 12. (a) | Explain Node.js and its event-driven architecture. Discuss modules, npm, HTTP modules, request-response cycle, asynchronous programming, and non-blocking I/O with examples. | CO4 | BL2 |
| 12. (b) | Develop a basic HTTP server using Node.js. The server should display different responses for at least two URL paths. Explain the working of the server, request object, response object, and status codes. | CO4 | BL3 |
| | OR | | |
| 13. (a) | Explain asynchronous file handling in Node.js. Demonstrate file creation, reading, writing, appending, and error handling using the File System module. | CO4 | BL3 |
| 13. (b) | Explain the EventEmitter class in Node.js. Develop a program to create custom events and handle them using listeners. Also explain the concept of middleware in web applications. | CO4 | BL3 |
| 14. (a) | Explain Express.js and its role in developing web applications. Discuss routing, GET and POST requests, middleware, custom middleware, serving static files, and request-response handling with examples. | CO5 | BL2 |
| 14. (b) | Develop an Express.js Student Management application with routes for displaying, adding, updating, and deleting student records using GET, POST, PUT, and DELETE methods. | CO5 | BL3 |
| | OR | | ------ |
| 15. (a) | Explain MongoDB as a NoSQL database. Discuss JSON and BSON, MongoDB data types, databases, collections, documents, and schema design with suitable examples. | CO5 | BL2 |
| 15. (b) | Write and explain MongoDB commands to create a database, create a collection, insert documents, retrieve documents, update documents, and delete documents for a Student Management System. | CO5 | BL3 |